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TransFuse: An Interpretable Transformer Ensemble Framework for Bangla Smishing Detection

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***Abstract*—The proliferation of mobile messaging has led to a surge in malicious activities like smishing, posing a significant threat to users, particularly in low-resource languages such as Bangla. While deep learning models have shown promise, their "black box" nature often hinders adoption in security-critical applications where trust is paramount. To address this, we introduce TransFuse, a robust and interpretable framework that employs a cross-validated ensemble of two state-of-the-art (SOTA) multilingual transformers, XLM-RoBERTa and mDeBERTa-v3, fine-tuned using a novel Layer-wise Learning Rate Decay (LLRD) strategy. On the publicly available BangalaBarta dataset, TransFuse achieves a new SOTA accuracy of 99.52%, demonstrating superior capability in classifying smishing, promotional, and normal SMS. Critically, we move beyond predictive accuracy by leveraging SHAP (SHapley Additive exPlanations) to provide faithful interpretations of the ensemble's decisions. This analysis confirms that our model's predictions are driven by semantically meaningful keywords, enhancing its trustworthiness. TransFuse represents a significant step towards building highly accurate and transparent AI systems for combating digital threats, offering a practical solution for securing mobile communications in under-resourced linguistic environments.**

***Index Terms*—Smishing Detection, Transformer Models, Ensemble Learning, Explainable AI (XAI), Cybersecurity**

I. Introduction

Short Message Service (SMS) forms part of one of the main modes of modern communication, but the ubiquity of the service is accompanied by the concomitant role of the method as a major carrier for malicious actors. Users are bombarded with unsolicited transmissions and one empirical investigation reports that within an interval of six months, American citizens were subjected to an astonishing 78 billion automated spam texts [1]. Such transmissions can vary from seemingly harmless promotional spam to dangerous smishing, or SMS-based phishing scams to support identity theft, fraud, and significant monetary loss [2]. The threat has now gone global as smishing attacks increased by 328% in one year, costing upwards of US$54.2 million [3]. This problem is especially sharp for languages like Bangala which feature relatively poor levels of documentation due to sparse annotated datasets and non-existence of purpose-built detection models used historically to mitigate these attacks [4].

To address this issue, research over the years has evolved from basic rule based filters to advanced machine learning techniques [5]. Initial deep learning based models such as ELMo, BERT etc has shown performance significantly outperforming the conventional spam classification approaches suited in the Bengali language [6]. In recent years, hybrid approaches that can deal with multiclass problems [6] and ensemble approaches that can further improve performance by using a voting and stacking algorithm have been proposed [7]. Despite these improvements two gaps remain, namely the application of ensembles based on modern multilingual transformer architectures is not explored enough and the "black box" nature of these models has been a big hindrance to the implementation of transparent and reliable security solutions.

In the light of the above problems, the selected objective of this study is to define a strong and transparent defensive strategy against smishing in Bangla language. The specific research objectives are:

- To construct a high-performance ensemble framework by combining XLM-RoBERTa and mDeBERTa-v3.
- To optimize model generalization on low-resource data using Layer-wise Learning Rate Decay (LLRD).
- To ensure trustworthiness in security applications by integrating SHAP for model interpretability.

To achieve these objectives, we introduce TransFuse. Our key contributions are:

- An ensemble of two SOTA multilingual transformers, XLM-RoBERTa [8] and mDeBERTav3 [9] was fine-tuned using Layer-wise Learning Rate Decay. Good adaptation to the domain of SMS requires (LLRD).
- Explainable integrating SHAP-based AI [10], providing. true execution of the decisions of the ensemble and Attainment of a new SOTA of 99.52% on the BangalaBarta

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dataset [11].

II. Related Works

Research on Bangla SMS classification has moved from the traditional machine learning approaches to deep learning and ensemble approaches. Khan et al. [6] were one of the first authors to move in this direction and introduce a novel Bangla SMS corpus and test a variety of models. Their results showed that ELMo is better than BERT with an accuracy of 94.02%, and also covers the contextual word representation in a better way. Simultaneously, research using traditional methods also yielded competitive results. Al Maruf [12] and colleagues presented the study of non deep learning ensemble methods of Bagging, Boosting, and Stacking using classical classifiers. The result emphasized the power of XGBoost The findings showed that carefully adapted traditional ensembles can be very powerful for binary spam detection.

Subsequently, the problem of multi-class distinguish smishing as well as promotional messages and ordinary SMS messages has been solved in sequential studies. Tanbhir et al. [13] proposed a hybrid architecture to combine the use of contextual embeddings computed from BERT and character level representations computed using a convolutional neural network. When tested on the dataset BangalaBarta, this way achieved an accuracy of 98.47%, in this way advocating the merits that are inherent in the fusion of the word- and character-level features. Building on ensemble methodology, Alvi et al. [7] presented a robust voting and stacking-based ensemble framework incorporating the complementary capabilities of Multinomial Naive Bayes classifier (MNB) and XGBoost classifier. Their system achieved an accuracy of 98.61 percent on a large-scale data set and thus demonstrated the effectiveness of meta-learning with an ensemble of base classifiers. Apart from these specific advancements, comparative studies like the ones conducted by Sharma et al. [14] evaluated a range of classical algorithms for which the results were consistent in supporting the suitability of MNB for Bangla text and in establishing good baseline performance figures against which future studies can further develop. Notwithstanding these substantive contributions, prior research has yet to study the contemporary multilingual transformer models or focus on the question of the interpretability of such models.

III. Methodology

The proposed TransFuse framework is based on a systematic, four-step pipeline to emsure both robustness and interpretability, as shown in Fig. 1. First, in the Data Preparation phase, we use stratified splitting and k-fold cross validation methods to ensure representing distribution of classes as well as unbiased testing the performance. Second, at the Base Model Fine-Tuning stage, we independently train XLM-RoBERTa and mDeBERTa-v3 instances following a Layer wise Learning Rate Decay (LLRD) learning strategy to successfully fine tune the pre-trained multilingual features for the particularities of Bangla SMS. Third, the Ensemble Aggregation stage is responsible for fusing these logits from the individual models, in order to reduce the variance of the individual models and improve generalization on data not seen by the models. Finally, the Model Interpretation phase uses SHAP to produce the token-level explanations, enabling the transparency and trustworthiness of decisions made by the system that can be applied to security applications.

A. Dataset and Experimental Setup

This study makes use of BanglaBarta corpus [11], a publicly available dataset which was designed particularly for identifying spam and smishing messages in Bengali language. The corpus contains 2,772 SMS messages that have been carefully classified into three different categories - Smishing, Promotional, and Normal. By using this dataset, the study brings out a realistic picture about the types of messages encountered by user in Bangladesh. Within the experimental framework, the data set was stratified initially to apportion an actual test set amount of 15% of the observations. This holdout subset was separated and then used during the final evaluation of the trained ensemble in order to ensure an unbiased evaluation of its performance on generalization. The remaining 85% of observations were classified as the development set. A 5-fold stratified cross-validation set up was used for the development set, to ensure the robustness of the model and reduce the dependency of the model on any individual random partition. Within each fold, eighty percent of the development data was used as the training set and the other twenty percent used as the validation set. Rather than using some form of aggressive text preprocessing, we used models known for learning contextual representations from raw text. The only required preprocessing was tokenization which was done using the respective tokenizers inherent in each model.

B. Model Architecture and Training Strategy

This paper focuses on assimilating two powerful pretraining multilingual models. Initially, basic architectures of these models are described, and then we explain their assimilation in our ensemble framework and describe the particular training regimes used.

C. Foundational Transformer Models

This paper focuses on assimilating two powerful pretraining multilingual models. Initially, basic architectures of these models are described, and then we explain their assimilation in our ensemble framework and describe the particular training regimes used.

XLM-RoBERTa (XLM-R): This model constitutes a cross lingual powerhouse based on the RoBERTa architecture [8]. Initially it was trained on a filtered set of CommonCrawl (2.5 TB) on 100 languages, including Bangla. Its major advantage is in its ability to learn shared representations among languages, and this makes it particularly suitable for learning languages that lack specific resources. In this study, we use the xlm-roberta-large.

mDeBERTa-v3: Multilingual mDeBERTaV3 [9] has two major innovations compared to previous models. It uses a

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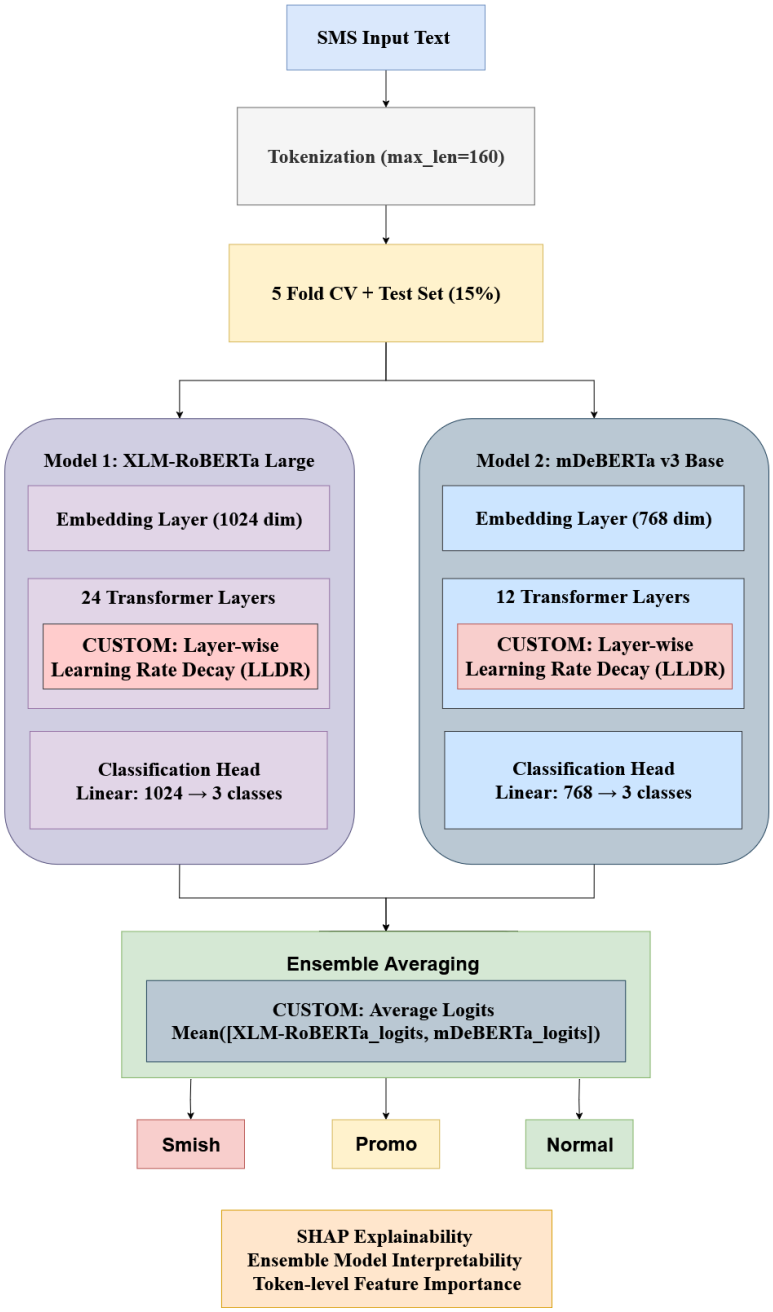


Fig. 1: The Proposed TransFuse Framework

disentangled attention mechanism, recruits content and relative position encoding, enhancing its word dependency modeling capabilities, first. Second, it is trained with an ELECTRA-style objective (Replaced Token Detection) rather than the Masked Language Modeling, which results in more sample efficiency and downstream performance. Our version is the mdeberta-v3-base version.

D. Cross-Validated Ensemble Construction

We build an ensemble by averaging in two stages to achieve the benefit of utilizing the rich representational forces of these architectures and to maximally improve generalization. To start with, on every base model (XLM-R and mDeBERTa), we will train five independent instances, each on one of the 5 cross-validation folds. The holdout test set predictions of these five instances are then averaged to produce a set of the output logits. This type of intra-model averaging results in a single robust predictor of every architecture.

Second, the final ensemble prediction is obtained by averaging the aggregated logits from both the robust XLM-R and mDeBERTa predictors. If $L_{M,f}(\mathbf{x})$ denotes the logits for input $\mathbf{x}$ from model architecture M trained on fold f, the final logits

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$L_{\text{ens}}(\mathbf{x})$ are given by:

$$L_{\text{ens}}(\mathbf{x}) = \frac{1}{2}\left(\frac{1}{5}\sum_{f=1}^{5} L_{\text{XLM-R},f}(\mathbf{x}) + \frac{1}{5}\sum_{f=1}^{5} L_{\text{mDeBERTa},f}(\mathbf{x})\right) \tag{1}$$

E. Layer-wise Learning Rate Decay (LLRD)

The use of Layer-wise Language Representation Distillation (LLRD) is an important part in our fine-tuning approach. The underlying rationale for this approach is that general language attributes are represented in the lower strata of a pre-trained model, while the upper strata represent more task-specific representations.

Varying learning rates are used throughout the depth of the model in order to retain critical knowledge and counteract catastrophic forgetting. The parameters are grouped together, and a baseline learning rate 1×10^{-5} has been set for the embedding layer. For each following transformer layer, the learning rate is gradually decayed by some multiplicative decay factor of $\eta = 0.9$. The top level classification head is given a different and higher learning rate of 2×10^{-5}. This stratified updating mechanism is conducive to more stable fine-tuning, which is more efficacious. Each model was trained for five epochs using the AdamW optimizer (weight decay 0.01). A sequence length of 160 tokens was used and an efficient batch size of 16 was preserved (made up of eight tokens per device and a two-step gradient accumulation scheme). The checkpoint with the best validation accuracy was taken as the best model for each fold.

F. Explainable AI for Model Interpretation

When the security across an application is critical, like smishing detection, it's important to understand the rationale behind a particular prediction of a model. To this end, we use SHAP [10], which is a game-theoretic technique that measures the marginal contribution of each input feature to the final prediction given by the model. It is interesting to note that SHAP analysis is performed on the final aggregated ensemble and not on each of the underlying base models. The prediction function that we have built is a simple or custom prediction encoder in which input text is passed into the encoder, tokenized, fed through all of the ten models that we have fitted based on the text, the logit averaging in a two stage process, and then the final class probability outputs are generated. This defines the function of the SHAP explainer in interpreting the ensemble's predictions on a level of individual tokens contained in the input SMS, thereby making a faithful representation of the whole system and giving the specific words that were vital for the classification of a message, so that the model will become transparent and more credible.

IV. Results and Discussion

In this section, the authors define the experimental setup, evaluate the performance of the proposed model, TransFuse, and compare it with the state-of-the-art baselines and analyze the decision-making process in the model framework by explainability techniques.

A. Experimental Setup

Experiments were performed on the platform, which had NVIDIA Tesla P100 along with 16GB VRAM. For the purpose of evaluating the real-time feasibility, the average inference latency was measured on the test set, which gave us 10.4ms per SMS message. Our model utilizes the PyTorch implementation, which makes extensive use of the Hugging Face Transformers and Datasets library for handling the model and data. With the help of scikit-learn, the performance metrics are calculated. The training process was conducted with the help of a well-chosen set of hyperparameters, which are presented in Table I. The learning rate and the configuration of the Layer-wise Learning Rate Decay schedule were chosen to ensure the stable convergence of the model, along with the generalized knowledge provided by the pre-trained models. This configuration was used across all 5 folds of the cross-validation process.

TABLE I: Key Hyperparameters for Fine-Tuning

Parameter	Value
Optimizer	AdamW
Epochs	5
Max Sequence Length	160
Train Batch Size (per device)	8
Encoder Base LR	1×10^{-5}
Decoder/Head LR	2×10^{-5}
LLRD Decay Rate (γ)	0.9
Weight Decay	0.01

B. Performance Analysis

The performance of the proposed ensemble model, which we name TransFuse, is outstanding in the task of Bangla SMS classification. The training process showed stable convergence, as demonstrated by the representative curves of the loss and accuracy plots for Fold 2, as presented in Fig. 2. For the XLMR and mDeBERTa models, the validation loss curve follows the training loss curve, and the validation accuracy curve levels off at a high value, which indicates that the proposed regularization and LLRD strategy are effective in preventing overfitting.

The final performance of TransFuse on the unseen holdout test set is summarized in the classification report in Table II. The model attained an overall accuracy of 99.52%. Precision, recall, and F1-scores remain consistently high across all three classes, with recall achieving perfect scores for the promo and normal categories, and near-perfect performance for the more challenging smish class. A finer-grained analysis is provided by the confusion matrix in Fig. 3. Out of 415 test samples, only two were misclassified. Notably, both errors involved the smish class: one smishing SMS was mistaken for a promotional message, and another was classified as normal. This suggests that the linguistic characteristics of some smishing messages can closely mimic benign or promotional content, representing the most challenging frontier for this classification task. Additional evidence of the model's discriminative capability

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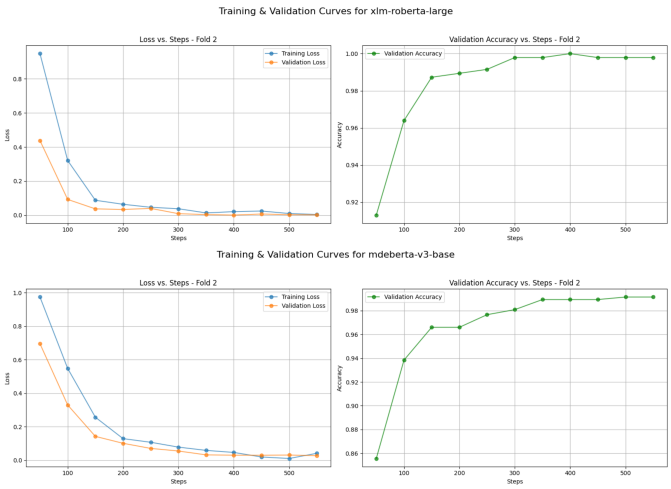


Fig. 2: Training and validation curves for XLM-RoBERTa and mDeBERTa on the best fold.

TABLE II: Classification report of TransFuse on the holdout test set.

Class	Precision (%)	Recall (%)	F1-score (%)
smish	100.00	98.55	99.27
promo	99.29	100.00	99.64
normal	99.29	100.00	99.64
Accuracy		**99.52**	
Macro Avg	99.52	99.52	99.52
Weighted Avg	99.52	99.52	99.52

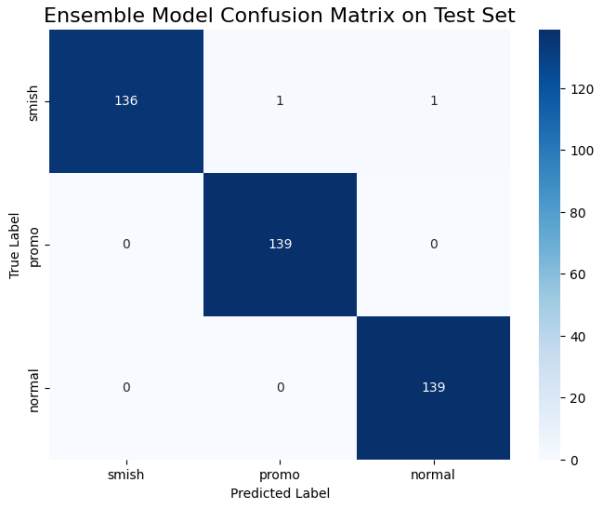


Fig. 3: Confusion matrix of the proposed model on the test set.

is illustrated in the multi-class ROC curve in Fig. 4. The Area Under the Curve (AUC) for each class is close to 1.00, and both the micro- and macro-average ROC curves are positioned near the top-left corner. This demonstrates an almost perfect ability to differentiate between classes across different decision thresholds.

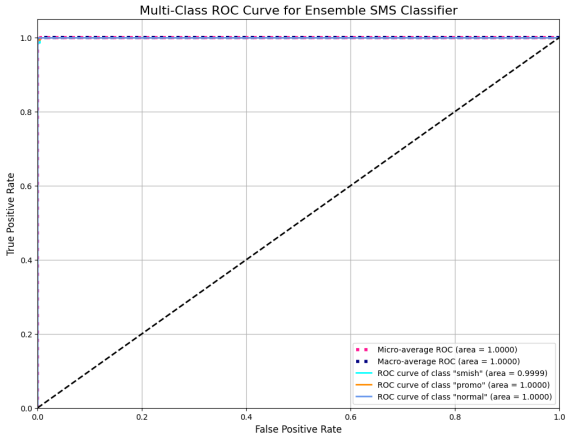


Fig. 4: Receiver Operating Characteristic (ROC) curve for the test set.

1) Comparison with Previous Methods: We compare TransFuse with several recent works in Bangla SMS classification. As shown in Table III, our method establishes a new SOTA, achieving an accuracy of 99.52%. Notably, on the same BangalaBarta dataset, our model surpasses the previous best result by Tanbhir *et al.* [13] by a margin of 1.05%. This significant improvement can be attributed to our robust cross-validated ensemble strategy, which combines the architectural strengths of two distinct, powerful transformer models, and our use of a tailored LLRD fine-tuning approach that optimizes task adaptation.

TABLE III: Performance Comparison with Previous Methods on Bangla SMS Classification

Reference	Year	Method	Accuracy (%)
Khan *et al.* [6]	2023	ELMo	94.02
Al Maruf *et al.* [12]	2023	ColonNet	96.31
Tanbhir *et al.* [13]	2024	CNN-BERT	98.47
Alvi *et al.* [7]	2025	Com(GS) Ensemble	98.61
Proposed	**2025**	**TransFuse (Ensemble)**	**99.52**

C. Interpretability with SHAP

In order to resolve the problem of the ”black box” nature of the deep learning model and ensure the presence of trustworthiness, we employed SHAP [10] to investigate the final ensemble model of TransFuse. SHAP provides the attribution of the prediction to the input tokens, which provides insights into the reasoning process. An example of the promotion message is presented in Fig. 5. It is evident that the selection of the model is dependent on the semantically significant keywords. It is also evident that the tokens related to the concepts such as package and discount are correctly identified as the key factors

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in the category of ”promo.” This validation of the logic of the model is essential for the practical application of the model in security.

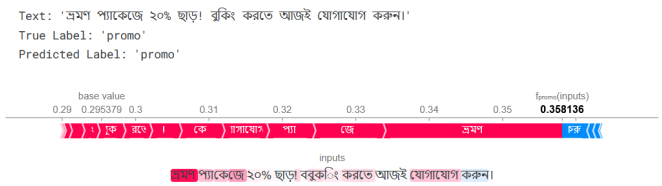


Fig. 5: SHAP analysis for a correctly classified promotional SMS, highlighting tokens contributing positively (red) or negatively (blue) to the promo prediction.

D. Future Research Directions

While TransFuse achieves SOTA performance, several avenues remain for future exploration and improvement. First, we plan to evaluate the model's robustness against adversarial attacks, such as character perturbation and homoglyph substitution, to ensure resilience in hostile environments. Second, to facilitate on-device deployment for real-time mobile protection, we will investigate model compression techniques like knowledge distillation and quantization to reduce the memory footprint without compromising accuracy. Finally, extending the framework to other low-resource Indic languages remains a key objective to broaden the system's applicability in diverse linguistic regions.

V. Conclusion

This paper presented a powerful and explanatory model of TransFuse, a Bangladeshi SMS classification system. We use a strategy that takes advantage of a cross-validated ensemble of two strong multilingual transformers, XLM-RoBERTa and mDeBERTa-v3, trained using a specialized Layer-wise Learning Rate Decay strategy. TransFuse sets a new SOTA on the BangalaBanta dataset with an excellent 99.52% accuracy and with better capabilities in detecting smishing, promotional, and normal messages. More importantly, we can go beyond the predictive accuracy and use SHAP to offer faithful interpretations of the decisions of the ensemble. This will resolve the most urgent problem of transparency of security applications, which will ensure that the classifications we make in our model rely on semantically meaningful linguistic properties and not spurious correlations. The blend of SOTA performance and inherent interpretability enables TransFuse to be an important leap in a new direction of creating more reliable and effective automated systems against digital threats in low-resource languages. Future research will be invested in applying this framework to other languages, comparing its resilience to adversarial smishing attacks, and considering model distillation as a way to implement it efficiently on the device.

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